This article has been accepted for publication in IEEE Network. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/MNET.2025.3576035

TeleQnA: A Benchmark Dataset to Assess Large Language Models Telecommunications Knowledge

Ali Maatouk*†, Fadhel Ayed*†, Nicola Piovesan†, Antonio De Domenico†, Merouane Debbah‡, Zhi-Quan Luo§
†Paris Research Center, Huawei Technologies, Boulogne-Billancourt, France
‡Khalifa University of Science and Technology, Abu Dhabi, UAE
§The Chinese University of Hong Kong, Shenzhen, China

***Abstract*—We introduce TeleQnA, the first benchmark dataset designed to evaluate the knowledge of Large Language Models (LLMs) in telecommunications. Comprising 10,000 questions and answers, this dataset draws from diverse sources, including standards and research manuscripts. This paper outlines the automated question generation framework responsible for creating this dataset and provides details on how we integrated human input at various stages to ensure the quality of the questions. Afterwards, using the provided dataset, an evaluation is conducted to assess the capabilities of several LLMs, including GPT-3.5, GPT-4, and Mixtral. The results highlight that these models struggle with complex standards-related questions but exhibit proficiency in addressing general telecom-related inquiries. Additionally, our results showcase how incorporating telecom knowledge context significantly enhances their performance, thus shedding light on the need for a specialized telecom foundation model. Finally, the dataset is shared with active telecom professionals, whose performance is subsequently benchmarked against that of the LLMs. The findings illustrate that LLMs can rival the performance of active professionals in telecom knowledge, thanks to their capacity to process vast amounts of information, underscoring the potential of LLMs within this domain. The dataset has been made publicly accessible on GitHub[1].**

I. Introduction

Large Language Models (LLMs) have catalyzed a revolutionary transformation in the domain of Natural Language Processing (NLP). By processing extensive corpora of text data, LLMs can discern patterns, predict outcomes, and craft coherent responses, setting new benchmarks for machine understanding and interaction [1]. The impact of LLMs extends far beyond the confines of NLP, permeating sectors as diverse as medicine [2] and finance [3], demonstrating the versatile applicability of these models. Much like these fields, the telecom industry stands on the brink of experiencing the potential impact LLMs could have on its landscape. Indeed, it is foreseen that LLMs may greatly facilitate a wide range of tasks, including troubleshooting anomalies, comprehending standards, elaboration of telecom chatbots, multi-modal forecasting, and providing optimization recommendations for network enhancement [4], [5].

The successful integration and adoption of LLMs within specific fields hinge critically on the models' depth of knowledge and expertise in those particular areas. To accurately assess and enhance this domain-specific knowledge, a robust mechanism for measurement is essential. Benchmark datasets emerge as the pivotal solution for this purpose [6], providing a structured framework to systematically assess the knowledge of LLMs in specialized domains. In the context of NLP, for instance, datasets like HellaSwag [7] and SIQA [8] have been tailored to test capabilities in commonsense reasoning and social intelligence, respectively. Similarly, in other areas such as medicine and finance, dedicated datasets like MultiMedQA [2] and FLUE [9] have been devised to evaluate the efficacy of LLMs in these fields.

As LLMs find their way into the telecommunications industry, a clear and pressing issue arises—there is a notable absence of a benchmark dataset designed to evaluate these models' proficiency in telecom. In fact, most current telecom datasets primarily consist of network information tailored for specific machine learning applications (e.g., network slicing [10]). Consequently, there is an urgent need for such a dataset, as highlighted in various prior research (e.g., [11]). This paper aims to bridge this gap by introducing TeleQnA, the first benchmark dataset tailored specifically for evaluating LLMs' telecom knowledge. The contributions of our paper are fourfold:

- First, we curate a comprehensive dataset that encompasses a vast corpus, spanning various disciplines within the telecommunications domain and taking on diverse forms, such as standards and research articles.
- Second, we introduce an automated Question and Answer (QnA) generation process based on the gathered material, involving two LLMs communicating with one another. We show how this architecture, complemented by human input at different stages, not only ensures the quality of the generated dataset but also facilitates its scalability to accommodate a wide array of multidisciplinary topics.
- Third, we proceed to assess the telecom knowledge of prominent LLMs such as GPT-3.5 and GPT-4 [12]. We illustrate their proficiency in responding to general telecom inquiries and highlight their deficiencies in addressing intricate questions related to standards specifications. Additionally, we demonstrate how the incorporation of contextual information significantly enhances these models' performance, particularly in areas where they typically face difficulties. All of these findings underscore the necessity for a specialized telecom foundation model to fully harness the potential of LLMs within the industry.
- Lastly, we assess the performance of professionals actively engaged in this field in comparison to that of LLMs. Our findings demonstrate that LLMs can compete with

*Equal contribution.
[1]https://github.com/netop-team/TeleQnA

Authorized licensed use limited to: Amrita Vishwa Vidyapeetham Chennai Campus. Downloaded on December 13,2025 at 02:37:27 UTC from IEEE Xplore. Restrictions apply.
© 2025 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Network. This is the author's version which has not been fully edited and
content may change prior to final publication. Citation information: DOI 10.1109/MNET.2025.3576035

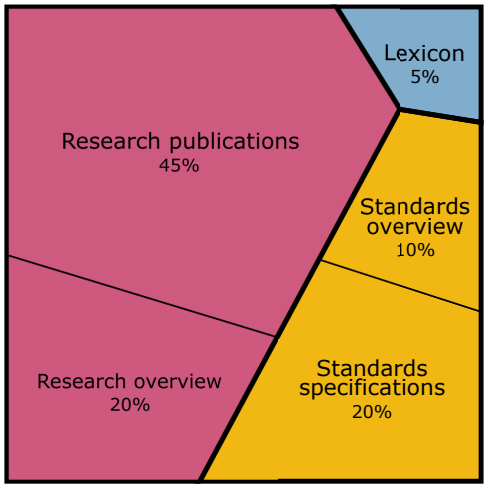


Figure 1: Distribution of the QnA dataset among the categories of the collected source materials.

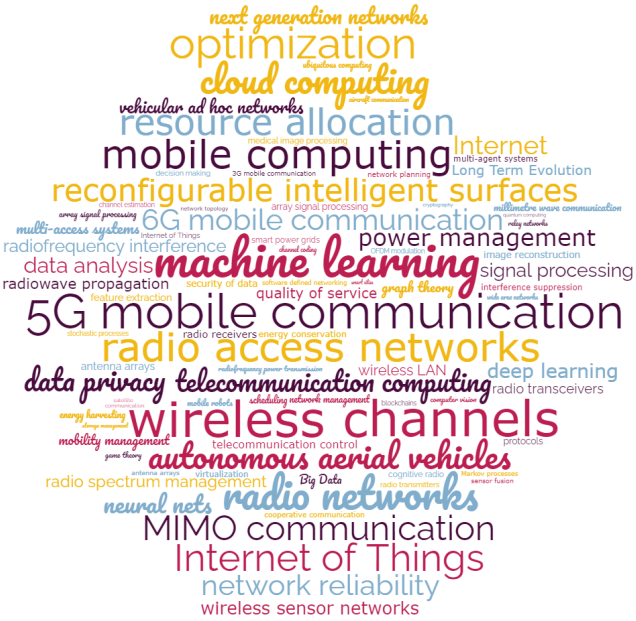


Figure 2: Word cloud illustrating the prominent keywords extracted from the collected research materials.

active professionals in telecom knowledge. This potential stems from their ability to process extensive volumes of information, highlighting the valuable role LLMs are poised to play in shaping the future of this domain.

Significantly, in addition to providing a dataset for evaluating the telecom knowledge of LLMs, this article introduces a methodology that extends across industries, offering universal applicability. This approach facilitates the creation of tailored evaluation datasets, enabling precise assessment of LLM capabilities in specialized domains.

II. Dataset Sources

When curating the dataset sources, our primary objective was to assemble a diverse and comprehensive collection of open-access data related to the telecom industry. This involved a broad spectrum of content, encompassing standards and scholarly research materials. In total, our collection comprised around 25,000 pages, containing approximately 6 million words. This approach guarantees that the resultant QnA dataset offers an in-depth and well-rounded depiction of the telecom industry's multifaceted aspects. Furthermore, this diversity serves as a valuable resource for evaluating the strengths of an LLM in various aspects, including standard specifications and general telecom expertise. The allocation of the QnA dataset among the three specified categories – research, standards, and lexicon – is illustrated in Fig. 1.

A. Standards Documents

Standards play a pivotal role in the field of telecommunications by ensuring that different technologies from various vendors can work together cohesively. These standards are established and maintained by recognized standardization bodies, including but not limited to 3GPP, IEEE, and ITU. Incorporating these documents when creating the QnA dataset is immensely valuable for capturing the intricacies of the telecom industry. Particularly, it allows delving deep into the technical facets and the present-day implementation of telecom technologies. With this in mind, we have integrated two distinct document sets into our raw data, all closely associated with standard activities.

- **Technical Specifications:** Technical specifications are detailed documents that define the technical standards for telecommunications systems. Given the sheer number of these documents, we have uniformly sampled a portion of them to ensure a non-biased representative selection of these documents across multiple standardization bodies.
- **Standards Overview:** To offer a more comprehensive perspective on standards, moving beyond the intricate technical specifications, we have gathered materials from various sources, encompassing standards summaries, review publications, and standards-related white papers.

B. Research Material

The landscape of publishing venues within the telecommunications field is vast. It encompasses a wide array of publishers, and is presented in diverse formats such as journals and conferences, with each repository housing a plethora of valuable information. The sheer abundance and diversity of material available pose a considerable challenge when attempting to create a QnA dataset based on these scholarly works. Additionally, the varying levels of citation and peer review standards, coupled with our commitment to exclusively use open-access material, introduce another layer of complexity to the task of curating a reliable and representative dataset. To address these challenges, we have categorized our sources into two groups: *Research publications* and *Research overview*. The former category primarily comprises technical details sourced from research articles and in-depth technical open-access books, while the second includes content from review

Authorized licensed use limited to: Amrita Vishwa Vidyapeetham Chennai Campus. Downloaded on December 13,2025 at 02:37:27 UTC from IEEE Xplore. Restrictions apply.
© 2025 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted,
but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Network. This is the author's version which has not been fully edited and
content may change prior to final publication. Citation information: DOI 10.1109/MNET.2025.3576035

and survey venues. Within each of these categories, we have adhered to the subsequent set of guiding principles:

- **Diversity:** Our first guiding principle was the diversity of material. In particular, we considered a broad spectrum of outlets, encompassing well-established venues hosted by various publishers.
- **Relevance:** Within these outlets, we have chosen to identify the most influential material by considering their citation counts over the past 15 years. Our selection of this time frame is deliberate, as it allows us to incorporate the most recent and pertinent findings. On the other hand, the number of citations serves as an indicator of a manuscript popularity and impact over time. To avoid bias toward older papers, which tend to accumulate more citations over time, we selected the top 100 most cited papers for each year. Furthermore, to infuse the dataset with the insights of industry experts, we have added material that telecom professionals have identified as pivotal contributions to this collection.
- **Bias Minimization:** To mitigate potential biases within the dataset, we have also taken a deliberate approach to uniformly sample a number of manuscripts from these venues. This strategy enhances the dataset diversity, encompassing not only popular topics but also those that, while less prominent, remain highly pertinent to the field.
- **Multi-disciplinarity:** We have ensured that our sources extend beyond the realm of pure communications and encompass topics relevant to the field. This expanded scope includes material from various domains, such as machine learning and optimization.

A word cloud representing the keywords extracted from the considered research material is illustrated in Fig. 2.

C. Telecom Lexicon

The telecommunications industry possesses a distinctive lexicon, characterized by an extensive array of technical terms, each holding significance in understanding the intricacies of communication networks. The incorporation of this lexicon into the generation of the QnA dataset holds significant value, as it provides a perfect avenue to evaluate the proficiency of an LLM in comprehending telecommunications terminology.

III. Dataset Characteristics

A. Question Type

With the source material prepared, our first course of action involved adopting a question type, and after careful consideration, we opted for a multiple-choice approach. This decision was made for several reasons:

- **Precision Assessment:** The multiple-choice format offers a straightforward means of gauging the accuracy of an LLM when presented with the dataset.
- **Complex Decision-Making:** The multiple-choice format allows us to probe deeper into the LLM's decision-making capabilities. It enables us to assess whether the LLM can accurately discern situations where multiple choices may be correct, rather than just one.
- **Nuanced Discrimination:** By adopting a multiple-choice framework, we can examine the LLM's ability to select the correct answer when presented with options that resemble the correct choice but are ultimately incorrect.

For the reasons mentioned above, multiple-choice approaches have gained popularity in the realm of machine learning benchmarks. For instance, they have been prominently utilized in datasets like MCTest [13].

B. Dataset Format

To maintain consistency across all questions, we have implemented a standardized format. Each question is represented in JSON format, comprising five distinct fields:

- **Question:** This field consists of a string that presents the question associated with a specific concept within the telecommunications domain.
- **Options:** This field comprises a set of strings representing the various answer options.
- **Answer:** This field contains a string that adheres to the format "option ID: Answer" and presents the correct response to the question. A single option is correct; however, options may include choices like "All of the Above" or "Both options 1 and 2".
- **Explanation:** This field encompasses a string that clarifies the reasoning behind the correct answer.
- **Category:** This field includes a label identifying the source category depicted in Fig. 1.

IV. Dataset Creation

To construct a comprehensive QnA dataset covering the multifaceted domain of telecommunications, a substantial number of questions is required. This task is further complicated by the specialized nature of telecom knowledge, demanding expertise to craft pertinent questions, answers, and explanations. Moreover, the telecom documents we collected often contain highly intricate information, making it infeasible for a team of human experts to generate questions and answers that comprehensively cover the diverse range of telecom subdomains. Adding to the complexity, we require these questions to be challenging to answer, and asking human experts to craft incorrect options that are also challenging across a wide array of subdomains is a task far from trivial. Given these challenges, adopting a crowdsourcing approach, similar to the one employed for the MCTest dataset [13], becomes impractical. To address these challenges, we leveraged OpenAI's GPT-3.5's APIto facilitate the generation process. This involved developing an automated framework based on LLMs, which will be elaborated on in this section. An overview of the entire process can be found in Figure 3.

A. Preprocessing

Our initial step involved cleaning and formatting the collected data sources for seamless integration with the utilized LLMs. This included removing irrelevant content such as pages with insufficient textual information (especially for semi-structured formats like PDFs), repetitive header and

Authorized licensed use limited to: Amrita Vishwa Vidyapeetham Chennai Campus. Downloaded on December 13,2025 at 02:37:27 UTC from IEEE Xplore. Restrictions apply.
© 2025 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted,
but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Network. This is the author's version which has not been fully edited and
content may change prior to final publication. Citation information: DOI 10.1109/MNET.2025.3576035

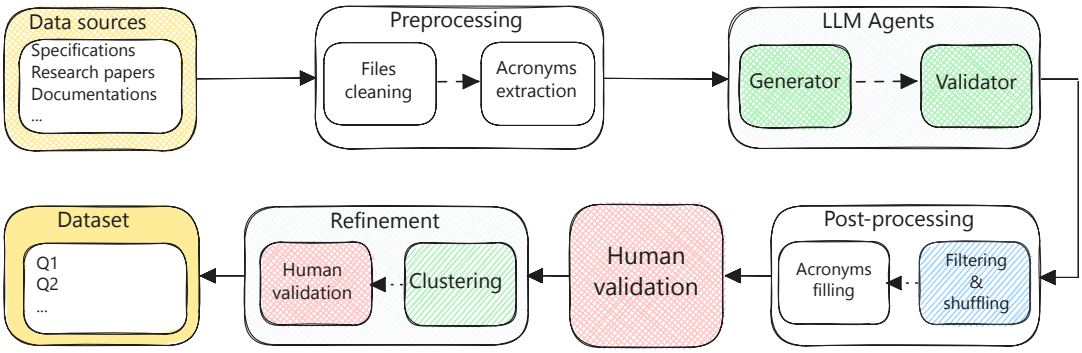


Figure 3: A high-level overview of the QnA generation process.

footer information, and non-textual elements like figures. In parallel, we created acronyms extractors that identify the acronyms found in these documents and their corresponding definitions, allowing us to grasp the terminology employed within these materials. After cleaning the data, we organized the content into semantically coherent chunks of a few thousand tokens, considering the token limitations of the models. These chunks were based on sections or subsections of the documents, with smaller, focused chunks improving the quality of the generation. This also allowed human annotators to select the most relevant sections more easily by using section titles to exclude less informative parts, such as appendices or bibliographies.

B. LLM Agents

Our LLM-based framework, inspired by Generative Adversarial Networks (GANs) [14], revolves around two GPT-3.5 LLM agents: a generator and a validator, each with distinct roles as outlined below.

- **Generator:** The generator is an LLM instance responsible for producing multiple-choice questions. To accomplish this, it is supplied with a segment of a telecommunications document, selected from those discussed in Section II. Subsequently, it is instructed to generate a set of multiple-choice questions based on the given context, adhering to the formatting guidelines. Furthermore, it is prompted to provide a rationale for the correct answer, typically by referencing sentences within the document to substantiate the choice of the answer.
- **Validator:** The questions and options generated by the generator, together with the contextual information (i.e., the relevant segment of the document used to generate the question), are given as input to another GPT-3.5 LLM, referred to as the validator. To prevent any biasing, the answer and the rationale behind the correct answer are intentionally omitted when providing input to the validator. The validator's primary function is to select an option among those provided solely based on its knowledge and the contextual information provided. If the validator picks the option designated as correct by the generator, the question is retained. However, if the validator selects an incorrect option, the question is discarded. This automated layer serves to enhance the overall accuracy and mitigate any potential instances of misinformation or fabrication that the generator may introduce.

C. Post-Processing

Moving forward in the process, we transition into the post-processing stage. Following the initial validation step, this phase focuses on the following key aspects:

- **Filtering:** We developed a tool designed to identify and filter out questions that are not self-contained, meaning they refer to the text from which are taken from by including particular terms such as "figure" and "fig".
- **Shuffling:** Our experiments have revealed a tendency of GPT-3.5 to place the correct answer in option 1. To mitigate this bias, we shuffled the options to ensure equal likelihood for all choices.
- **Acronyms:** We designed acronym detectors to identify acronyms in the questions and map them to their respective definitions, leveraging LaTeX's dedicated acronym commands and the table of abbreviations and acronyms typically found in standards documents..

D. Human Validation - First Stage

Having validated the questions using the validator and completed the necessary post-processing steps, we now introduce the first stage that incorporates human intervention. Specifically, telecom experts are assigned the task of carefully reviewing each question, the provided answer options, the explanation given by the question generator for the answer, and the contextual information used. This step is crucial because the generator and validator may occasionally fail to capture nuanced meanings, leading to incorrect answers, or they may generate overly specific questions that lack general relevance — such as asking about a specific figure, which would not be available to the LLM under test — while our focus is on evaluating broader telecom knowledge. Although our pipeline is designed to minimize such issues, human validation remains essential for filtering out these cases.

The goal of the human validation phase is to make an informed decision on whether to retain or discard a question. This human-in-the-loop process serves two primary objectives:

- **Ensure question correctness:** The expert's role is to verify the accuracy of the questions, their associated

Authorized licensed use limited to: Amrita Vishwa Vidyapeetham Chennai Campus. Downloaded on December 13,2025 at 02:37:27 UTC from IEEE Xplore. Restrictions apply.
© 2025 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted,
but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Network. This is the author's version which has not been fully edited and
content may change prior to final publication. Citation information: DOI 10.1109/MNET.2025.3576035

	Run 1	Run 2	Run 3	Run 4	Mean	Std
50 questions	63.35	65.63	65.10	63.79	64.46	0.92
25 questions	65.19	67.56	66.90	66.30	66.48	0.87
10 questions	67.64	64.32	66.94	67.20	66.52	1.29
5 questions	65.45	67.55	68.52	66.85	67.09	1.11
1 question	66.43	67.48	68.10	66.46	67.11	0.70

Table I: Illustration of GPT-3.5's accuracy (%) across multiple runs and question batch sizes.

options, and the designated correct answer, effectively eliminating improperly generated questions.

- **Ensure that the questions are self-contained:** The questions should not revolve around content that is of localized interest, such as experimental results specific to a proposed algorithm.

E. Refinement

The collected telecom sources inherently exhibit overlaps across the various venues. This redundancy leads to the presence of overlapping questions within the QnA dataset, presenting a challenge that calls for meticulous dataset refinement. To address this issue, we utilized OpenAI's Ada v2 text embeddings model, which facilitates the conversion of all questions and their corresponding explanations in the QnA dataset into text embeddings. Crucially, these vectorized representations enable us to assess the semantic similarity between pairs of questions by evaluating the distance between their embeddings in a multidimensional space. Upon generating these embeddings, we implemented the K-Means clustering algorithm. This clustering serves as a foundational step for streamlining the refinement process, setting the stage for more focused and efficient human review and intervention. In fact, the concluding phase encompasses a secondary human validation that serves a dual purpose:

- **Eliminating redundant questions:** The primary goal of the intervention is to eliminate duplicate or semantically-redundant questions within each cluster.
- **Final iteration of quality checking:** The second objective entails conducting a final round of verification, similar to the first human validation stage.

V. Performance Evaluation

In this section, we conduct a comprehensive evaluation of GPT-3.5 and GPT-4 using the TeleQnA dataset. Our objective is to assess their proficiency in telecom knowledge while also offering insights into their strengths and weaknesses. Additionally, we compare the performance of telecom professionals to these language models, establishing a benchmark for reference. As a performance measure, we define the accuracy as the percentage of questions for which the entity at hand selected the option marked as correct in the dataset.

A. Batch Size and Variance Study

Our numerical analysis begins by examining the variance of the accuracy achieved by the LLMs when responding to the TeleQnA dataset. Specifically, we are interested in tracking how the accuracy evolves with each iteration of LLM questioning. Additionally, we investigate the impact of sending questions in random batches of varying sizes ($B = 1, 5, 10, 25,$ and 50) to the LLM. The former investigation aims to show the number of iterations required to obtain a statistically reliable accuracy score for the LLM's telecom knowledge. Meanwhile, the latter investigation seeks to identify trends when querying the LLM's API with larger question batches, potentially streamlining the process by sending questions in batches rather than one by one. To obtain these findings, we uniformly sampled 1000 questions and conducted accuracy assessments across multiple runs and with different batch sizes. The results of these experiments are presented in Table I for GPT-3.5[2].

The results demonstrate the consistency of accuracy across multiple runs, with a standard deviation of approximately 1%. Given that the standard deviation decreases with the number of questions considered, it can be inferred that this deviation is even smaller when applied to the entire dataset. Therefore, we can conclude that conducting a single run is sufficiently reliable for assessing the proficiency of the LLM. Conversely, we observe a decline in mean accuracy when questions are submitted in larger batches. This decrease may be attributed to the inherent diversity of questions stemming from various sub-domains within telecommunications. When questions are batched together, the LLM encounters a broader spectrum of topics, which may result in less specialized responses and, therefore, a potential drop in accuracy. However, we note that this decline in performance remains modest when transitioning from a batch size of $B = 1$ to $B = 5$. Thus, we have chosen to utilize this batch size in the rest of our experiments to reduce the number of queries needed for assessment.

B. Performance Benchmarks

We evaluate the performance of GPT-3.5 and GPT-4 across the various categories of the TeleQnA dataset. For comparison, we also evaluate the performance of two open source models, a small model Mistral 7b, and a medium model, Mixtral 8x7B mixture of experts (MoE). The latter is a mixture of experts model created by Mistral, consisting of 8 experts each with 7 billion parameters, totaling 56 billion parameters. At every layer of Mixtral, for every token, a router network chooses two of the 8 experts to process the token and combine their output.

The results are reported in Table II. As anticipated, GPT-4 consistently outperforms GPT-3.5, demonstrating around 7% improvement across all categories. We remark that the open-source alternative Mixtral MoE performs similarly to GPT-3.5, while Mistral 7b significantly underperforms, which is expected given its small size. Notably, LLMs exhibit exceptional performance in the lexicon category, which encompasses general telecom knowledge and terminology, achieving approximately 87% accuracy for GPT-4. Such discrepancies in performance may stem from multiple factors. Larger models, with their greater number of parameters, are able to capture more intricate patterns in the data, leading to improved performance. In addition, the quality and diversity of the data

[2]GPT-4 exhibited similar trends, and therefore, we have opted to exclude the specific details.

Authorized licensed use limited to: Amrita Vishwa Vidyapeetham Chennai Campus. Downloaded on December 13,2025 at 02:37:27 UTC from IEEE Xplore. Restrictions apply.
© 2025 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted,
but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Network. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/MNET.2025.3576035

	Mistral 7b	Mixtral (MoE)	GPT-3.5	GPT-4	Humans
Lexicon (500)	56.80	83.80	82.20	86.80	80.33
Research overview (2000)	51.60	70.70	68.50	76.25	63.66
Research publications (4500)	49.27	70.20	70.42	77.62	68.33
Standards overview (1000)	46.20	66.73	64.00	74.40	61.66
Standards specifications (2000)	35.60	55.85	56.97	64.78	56.33
Overall accuracy (10000)	47.07	67.74	67.29	74.91	64.86

Table II: Illustration of Mistral 7b, Mixtral (MoE), GPT-3.5, GPT-4, and active professionals accuracy (%) across the various TeleQnA categories.

used to train the models play a crucial role in determining their performance. Post-training refinement techniques further influence the ability of the model to handle specialized tasks such as question answering.

Besides, the models face challenges when confronted with more intricate questions related to standards, with the highest performing model, GPT-4, achieving a modest 64% accuracy in this domain. To explain this limited performance, let us consider a question asking the physical resource block (PRB) number in a 5G cell with a band of 10 MHz and a subcarrier spacing of 15 kHz; a vanilla LLM typically fails to provide the right answer, as this does not only require to perform a simple computation involving the provided input and the number of subcarriers in a PRB but also to know the size of the guardband for the specific bandwidth and subcarrier spacing configuration. This requires the LLMs to capture the dependency of different pieces of information – often included in tables rather than plain text – provided in standard documents.

In summary, GPT-3.5 averaged an accuracy of 67%, while GPT-4 achieved an accuracy of 74%. These results demonstrate that these models possess a solid foundation in general telecom expertise. However, to attain higher accuracy in responding to complex inquiries, further adaptations to the telecom domain are necessary.

In our final step, we conducted a performance benchmark comparing active professionals to LLMs. Fifty professionals were provided with a set of 30 questions and asked to answer them without access to any external material. Questions were sampled from the dataset, allowing overlap between professionals to improve the consistency of answers. Given the number of professionals involved and the number of questions answered by each of them, the results were statistically consistent, indicating that LLMs and active professionals perform comparably in general telecommunications knowledge. However, in the case of intricate questions related to research and standards, LLMs demonstrate the capability to rival these professionals. This is attributed to LLMs' ability to digest and memorize complex and intricate documents. Furthermore, it is crucial to recognize the challenge faced by professionals when responding to these questions, as they encompass a broad range of telecom subdomains that these individuals may not be necessarily actively engaged with in their work. Considering all factors, our results underscore the significant promise that LLMs hold within this domain, as demonstrated by their competitiveness within this extensive and comprehensive dataset.

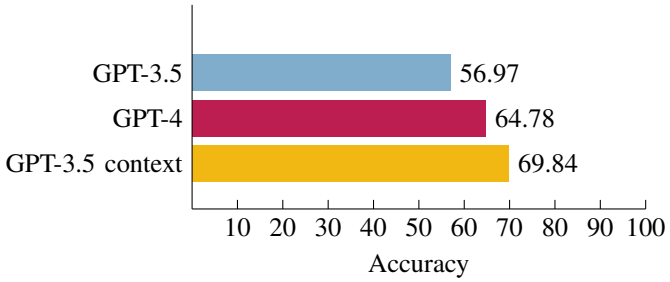


Figure 4: Accuracy (%) comparison among GPT-3.5, GPT-4, and GPT-3.5 with context in the standards specifications category.

C. Influence of Context

Until this stage of our benchmarking process, we have been querying the LLMs without accompanying contextual information for the questions. Nevertheless, in this subsection, our goal is to investigate how supplementing questions with additional context affects the accuracy of these models.

To accomplish this, we have focused on the standards specifications sources, encompassing thousands of technical standards pages. Our selection of this category is driven by the fact that this is where LLMs have exhibited the lowest performance. With this in mind, we segmented these pages into approximately 500-word segments before generating embeddings for each segment using OpenAI's Ada v2 text embeddings. Moreover, we employed the same OpenAI model to create embeddings for the questions and corresponding options belonging to this category. Following that, we constructed a distance matrix between the embeddings of each question-options pair and those of each segment. The next step involved querying the LLM by supplying batches of five questions-options pairs, and additionally, as context, the top-3 closest segments to the five question-option pairs based on the distance matrix. The outcomes of these experiments are illustrated in Fig. 4.

GPT-3.5 supplemented with contextual information reached an accuracy level of 69.84%,[3] which translates into a relative accuracy enhancement of 22.5% compared to the scenario lacking context. This gain highlights the significant enhancement in performance achieved by incorporating contextual information, demonstrating how even less advanced models like GPT-3.5 and Mixtral can match the performance of state-of-the-art GPT-4 model. This underscores the necessity for a specialized telecom language models, fine-tuned on standard specifications and public documents from vendors and service providers.. Developing such a foundation model has the potential to push the boundaries of LLM performance in the telecom domain, paving the way for a wide range of use cases that demand telecom knowledge and expertise. Moreover, in

[3]The context-enhanced version of Mixtral achieved a score of 70.43%.

Authorized licensed use limited to: Amrita Vishwa Vidyapeetham Chennai Campus. Downloaded on December 13,2025 at 02:37:27 UTC from IEEE Xplore. Restrictions apply.
© 2025 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Network. This is the author's version which has not been fully edited and
content may change prior to final publication. Citation information: DOI 10.1109/MNET.2025.3576035

real-world telecom applications, such as network optimization or fault detection, contextual enhancement through Retrieval Augmented Generation (RAG) could be operationalized by integrating the model with real-time network data, operational logs, and external knowledge bases [15]. This would enable the model to provide answers tailored to the current state of the network, offering more actionable insights for decision-making.

VI. Limitations

A. Human subjectivity

While our framework's design is aimed at minimizing errors in the questions, it remains dependent on human judgment to determine whether a generated question should be retained or discarded. This introduces an element of subjectivity, particularly in cases where questions offer multiple valid options, albeit with varying degrees of correctness. In such instances, human subjectivity comes into play. For instance, let us consider the following question:

Question 1: What advantages does a data center-enabled HAP (High Altitude Platform) offer from an energy perspective?

- *Option 1:* Saves cooling energy and harvests solar energy
- *Option 2:* Uses renewable energy and offsets carbon emissions
- *Option 3:* Deploys large-scale solar panels and batteries
- *Option 4:* Requires less energy for communication links

Answer: *Option 1:* Saves cooling energy and harvests solar energy

Explanation: The data center-enabled HAP saves cooling energy due to its location at the naturally low temperature stratosphere and harvests solar energy using large solar panels.

Category: Research publications

An expert might discern that Option 1 is more accurate than Option 2. In fact, although Option 2 is indeed an advantage offered by data center-enabled HAP, Option 1 provides more insights into the fact that high-altitude platforms are deployed in the low temperature stratosphere, while also considering the fact that solar energy is used (i.e., a renewable energy). We eliminated some of these questions to ensure the utmost objectivity and correctness, but deliberately retained others to challenge the LLM and assess its capability to discern between options exhibiting differing degrees of correctness.

B. Format and scope

For TeleQnA, we chose MCQs for their clarity and ease of scoring. The challenge with open-ended questions lies in the difficulty of assessing answers, often requiring an LLM as a judge, which can introduce scoring inconsistencies. However, open-ended questions are valuable for capturing more nuanced aspects of model performance. In future work, we plan to incorporate these questions to evaluate not only factual knowledge but also reasoning, critical thinking, and problem-solving, providing a more comprehensive assessment of LLM performance.

VII. Conclusions

In this paper, we introduced TeleQnA, the first open-source benchmark dataset tailored specifically to the telecom industry. With a collection of 10,000 questions and answers drawn from various sources in the field, TeleQnA serves as an evaluation tool for assessing the knowledge of LLMs in the telecommunications domain. Throughout this paper, we have detailed the development process of TeleQnA and conducted an extensive evaluation of GPT-3.5 and GPT-4 using TeleQnA. The results have revealed interesting insights: although these models excel in addressing general telecom-related questions, they face challenges when tackling complex inquiries. Importantly, our findings emphasize the critical need for specialized telecom foundation models.

References

[1] T. Kojima *et al.*, "Large language models are zero-shot reasoners," in *Advances in Neural Information Processing Systems*, 2022.
[2] K. Singhal *et al.*, "Large language models encode clinical knowledge," *Nature*, vol. 620, no. 7972, pp. 172–180, Aug. 2023.
[3] S. Wu *et al.*, "BloombergGPT: A Large Language Model for Finance," *arXiv preprint arXiv:2303.17564*, May 2023.
[4] A. Maatouk *et al.*, "Large language models for telecom: Forthcoming impact on the industry," *arXiv preprint arXiv:2308.06013*, August 2023.
[5] L. Bariah *et al.*, "Understanding Telecom Language Through Large Language Models," *arXiv preprint arXiv:2306.07933*, 2023.
[6] Y. Liu *et al.*, "Datasets for large language models: A comprehensive survey," 2024. [Online]. Available: https://arxiv.org/abs/2402.18041
[7] R. Zellers *et al.*, "HellaSwag: Can a machine really finish your sentence?" in *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, Jul. 2019, pp. 4791–4800.
[8] M. Sap *et al.*, "Social IQa: Commonsense reasoning about social interactions," in *EMNLP-IJCNLP 2019*, Nov. 2019, pp. 4463–4473.
[9] R. Shah *et al.*, "When FLUE meets FLANG: Benchmarks and large pretrained language model for financial domain," in *EMNLP 2022*, Dec. 2022, pp. 2322–2335.
[10] A. Thantharate *et al.*, "Crawdad umkc/networkslicing5g," 2022. [Online]. Available: https://dx.doi.org/10.15783/k0w0-js18
[11] M. Kotaru, "Adapting foundation models for information synthesis of wireless communication specifications," *arXiv preprint arXiv:2308.04033*, August 2023.
[12] OpenAI, "GPT-4 Technical Report," *arXiv preprint arXiv:2303.08774*, March 2023.
[13] M. Richardson, "Mctest: A challenge dataset for the open-domain machine comprehension of text," in *EMNLP 2013*, October 2013.
[14] I. Goodfellow *et al.*, "Generative adversarial networks," *Commun. ACM*, vol. 63, no. 11, p. 139–144, oct 2020.
[15] A.-L. Bornea *et al.*, "Telco-rag: Navigating the challenges of retrieval-augmented language models for telecommunications," in *Proceedings of the IEEE Globecom*, December 2024.

Authorized licensed use limited to: Amrita Vishwa Vidyapeetham Chennai Campus. Downloaded on December 13,2025 at 02:37:27 UTC from IEEE Xplore. Restrictions apply.
© 2025 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted,
but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.